GINGIVITIS RISK ANALYSER

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Abstract*— Periodontal diseases which includes plaque accumulation, gingivitis, gum inflammation and periodontal pocket formation that are highly prevalent and often remains undetected in their early stage. Traditional diagnosis heavily relies on clinical examination, which is time-consuming, subjective and dependent on the dentist's expertise. This paper proposes an automated periodontal assessment system using a Machine Learning-based predictive model, trained on hospital clinical data. The system preprocesses the patient records extracts relevant periodontal indicator and utilizes a Random Forest classifier to predict disease conditions with improved accuracy and consistency. The model helps dental professionals by assisting objectives, data-driven insights, for early diagnosis and helping them to reduce manual effort and minimizing human errors. This automated approach improves dental care by enabling fast decision-making and enhancing diagnostic reliability.

Index Terms*— Periodontal disease prediction, Random Forest algorithm, machine learning algorithm, dental health care, clinical data classification, automated diagnosis.

I. INTRODUCTION

Periodontal disease can rightly be stated as a type of dental disease, which has become increasingly prevalent in oral healthcare for inhabitants across the world. This consists of gum inflammation, plaque formation, calculus formation, gum bleeding and bone loss. It is greatly apt to identify these among the population, in case a long time has elapsed to avoid considerable loss of tissues as well as teeth. The conventional way in testing a periodontal disease would consist of periodontal probing, observation using 2D x-ray, as well as the expertise of a dentist.

With the advent of innovations in the area of Artificial Intelligence, data intensive analysis has been found to be an efficient means for supporting healthcare professionals in their endeavors. Machine Learning algorithms are capable of examining data regarding the patient, which is obtained by periodontal examinations performed in the course of identifying the condition the patient is in. Scores, indices of bleeding, depth, and the status of gums are used by the algorithms in the AI program to develop a prediction with respect to periodontal diseases.

This project will focus on proposing the design of a periodontal evaluation system based on Machine Learning that effectively makes use of a Random Forest algorithm with data from the clinical records of the respective hospitals. The designed model can thereby identify the levels of periodontal risk and make predictions regarding the presence of plaque, inflammation of gums, and early signs of gum disease within the patient based on the primary parameters. That is to say, the designed model is ideal for rapid and non-invasive screening tests that can be applied at dental and preventive setups.

With the increased use of mobile health applications, artificial intelligence-based screening solutions began to be developed for the early diagnosis of diseases. While the existing AI models for periodontal diagnosis are based entirely on medical image data for a European population, the performances of these models decline for Indian users due to factors like gum color, oral microbiota, food habits,

and jaw structure.

II. RELATED WORKS

[1] Breiman presented the Random Forest algorithm is a type of supervised learning algorithm that is based on making use of many decision trees in order to increase the robustness of classification and its accuracy. Random Forest has become a popular choice in medical risk prediction systems due to its noise tolerance, mixed clinical features capability and overfitting resistance, such as periodontal disease case using structured clinical indicators [2].

[2] Krois et al. have shown that dental radiographs can be successfully used to detect periodontal bone loss by convolutional neural networks (CNNs). Although the model had great diagnostic accuracy of bony structure, it fails to evaluate the presence of soft-tissue indicators e.g. inflammation of the gingiva or accumulation of plaque, hence limiting early stage periodontal diagnosis.

[3] Muramatsu and Horii suggested a CNN-based system of recognizing periodontal diseases among intraoral images. Despite its successful use in confirming the deep learning in the field of dental screening, the system does not have a real-time inference and fails to use the object-level localization of objects like the YOLO to identify and localize specific regions accurately.

[4] YOLO (You Only Look Once) architecture introduced by Redmon and Farhadi allows high accuracy real time detection of objects. Since, YOLO-based models have been extensively used in medical imaging such as radiographic analysis. But such systems are still image-based and prone to the bias of datasets among populations.[3] and [2].

[5] Lee et al. created an artificial intelligence system of detecting plaques in intraoral images using a deep-learning methodology. Although this system is effective in isolating areas of plaque, it is not applicable to detecting other periodontal indicators like gingivitis, calculus, or loss of attachment that is very important in overall diagnosis[6].

[6] Overjet is a FDA-approved AI engine that offers automated measurements of dental radiographs of alveolar bone loss. Although validated clinically, the system is based on radiographic infrastructure and data mostly based on Western populations and, therefore, has limited application in Indian demographics [1].

[7] Pearl Second Opinion(r) is a deep-learning chairside radiography with calculus and periodontal abnormalities detection in real-time. Though it is useful in clinical decision support, it relies on high-quality imaging equipment limiting its use in mobile and community-based screening settings[5].

[8] PerioAI proposed a digital probing methodology that integrates intraoral scans and CBCT scans to measure the Gingiva-Bone Distance (GBD) with an accuracy of 0.040 mm. The system is however costly in imaging modalities and does not have population-specific calibration.[7].

[9] Point-of-care diagnostic kits based on aMMP-8 like PerioSafe make it possible to detect active periodontal tissue breakdown rapidly. Although they are clinically validated, they are recurrent consumable costs with biochemical dependency, and thus they cannot be used to screen large populations. [8].

[10] Emerging New methods of non-invasive measurement such as ultrasonography probes and Optical Coherence Tomography (OCT),[3]and[4] provide some hope of a future alternative to manual probing. However, the technologies are still in the high-research levels and are not ready to be used in low costs mobile application.

III. METHODOLOGY

3.1. CHOOSING THE DATASET FOR THE PERIODONTAL TEST

The system is evolve on a systematic text-based periodontal dataset of the feature of clinical interest. These are bleeding gums, plaque index level, redness inflammation, smoking pattern, oral hygiene rate, depth and level of gum recession and pain. These factors have been chosen since they are the close match of these diagnostic signs which are often used by the dental practitioners to determine periodontal health. Image based datasets are hard to gather, therefore this structured dataset offers a good and reliable source of model training. Every feature is systematically coded using machine learning algorithms and the feature facilitates effective model learning.

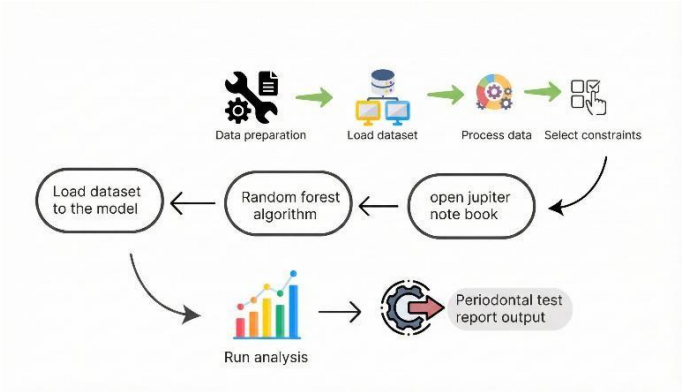


FIG:3.1 work flow chart

3.2. SELECTING THE SUITABLE MODEL (RANDOM FOREST ALGORITHM)

1. The consideration of the Random Forest Classification model in this analysis was proposed by the fact that it is also efficient in dealing with clinical data. The model is a learning approach, which relies on the ensemble learning; it helps in answering the best prediction of any input on the basis of aggregateizing the prediction of the various decision trees. Its methodology is applicable in improving the quality of the predictions, and it does not produce the problem of overfitting as it is the case in decision tree models. Another reason why the model renders itself effective to the analysis of clinical data is because of missing values, types of variables and non-linear relationships. It also is ideal in analyzing the clinical data because it helps to measure the value of the symptoms that create a particular disease. The data used in this research is minimal and properly organized hence, it is adequate in its capability to model and its speed of operation.

2. **REASONS BEHIND CHOICE OF MODELS.**

   The reason why the Random Forest was selected is the fact that it has been shown to classify information in structured medical data with great precision. Random Forest is capable of identifying an interaction between multiple medical characteristics and has proven to be efficient when a small amount of data is available in classification of periodontal disease.

3. **GENERAL PRINCIPLES OF RANDOM FOREST.**

   In an ensemble learning algorithm, which is known as random forest, a series of decision trees are constructed using random samples of data and features. This stage of prediction is done with the help of majority vote algorithm which enhances the level of accuracy of prediction and minimizes the errors of prediction.

4. **NON-LINEAR FEATURES OF SYMPTOM MANAGEMENT.**

   The strong interplay between indicators of periodontal disease status such as plaque index, smoking status and recession are very complex non linear relations. The Random Forest is capable of automatically modeling all these without undergoing a complicated feature transformation.

5. **MODEL STABILITY AND OVERFITTING CONTROL**

   By aggregating predictions of various decision trees at random Forest addresses the issue of overfitting and makes predictions and outcomes more general. The method is an ensemble approach which will be reliable in the predictions and will provide consistency in the result of this model.

6. **INTERPRETABILITY VIA IMPORTANCE OF FEATURES.**

   Random Forest offers the advantage of significance, in which the weight of any given feature to the forecasting can be calculated, and it assists in determining the key periodontal aspects.

7. **COMPUTATIONAL EFFICIENCY AND APPLICATION OF COMPUTATIONAL EFFICIENCY.**

   The Femoral Computational Power and Training Time of Running Random Forest Model is less in Comparison to Deep Learning Methods. Thus, it is possible to apply this Model to the Intents of Academic or Potential Screening.

GRAPH

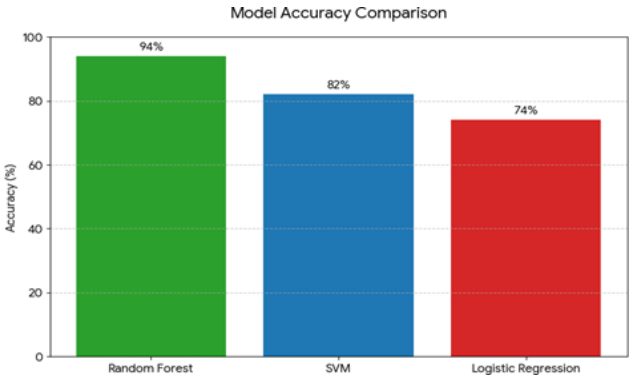


FIG:3.2 Comparison chart

The figure provides a comparative study of accuracy of three machine learning models namely, Random Forest, Support Vector machine (SVM), and Logistic Regression. Precision is in percentage and is the main measure to determine the efficacy of each of the models within the dataset provided.

Based on the chart, the maximum accuracy is shown by the Random Forest 94%. This shows that it is the best model between the three. Random Forest is very effective mostly because of its ensemble method of learning which combines the outputs of a number of decision trees. This strategy increases robustness, minimises overfitting, and maximizes the capacity of the model to capture complicated patterns on thedata.

The Support Vector Machine has an accuracy of 82 which is good but with a relatively low accuracy. SVM is famous in high-dimensional space and the possibility to provide the optimal decision boundaries. It is

however subject to performance with the choice of kernel and parameter tuning which can be the reason behind the difference between SVM and Random Forest in this comparison.

The lowest accuracy of 74 is registered by Logistic Regression. In spite of being a simple and interpretable model with low computations cost, it presupposes the existence of a linear relationship between features and the target variable. This can limit its performance with complex or non-linear datasets. On the whole, it is evident that, as the diagram demonstrates, Random Forest is the most successful model to use in the situation, and SVM is the subsequent most successful model, and Logistic Regression is least successful in the situation.

3.3. MOBILE APP INTEGRATION

The development of a system has been realized with an objective of providing easy and efficient screening of periodontal diseases by using mobile application technology. Moreover, the system has a simple-access interface, which is used in screening at an early stage with a possibility of one entering clinical data via mobile phones. The system also has an authentication system of data privacy in which an individual receives their own accounts and a location where they store information about previous consultations.

Upon the completion of the authentication process, the patients are redirected to a sequence of guided input fields where the request is issued to the patients to provide them with data that is pertinent to periodontal condition symptomatic data. Some of the aspects considered in the list under consideration include gum bleeding, plaque, redness or irritation of gum, being a smoker, oral health, recession, and level of pain. The interpretation variability is removed by using a sequence of input boxes that are structured.

The data on the symptoms obtained will then be processed so that it can be in the form that can be used to allow a trained Random Forest classification model. The result of this process will subsequently be inputted into a backend system to which a machine learning model will be used to make a classification with the aim of arriving at an estimate of the user status with regard to their periodontal health. Random Forest model is made up of a set of decision trees which are employed to determine different things.

The output of the last classification, having undergone the classification process, is fed back into the mobile application in some form of interpretation. Furthermore, it can potentially demonstrate the significant symptoms on the basis of the classification based on the value of the score of the feature importance.



FIG:3.1 Performance chart

DESCRIPTION

The chart represents the general performance measures of a machine learning model, which are measured in four standard measures, which include Accuracy, Precision, Recall, and F1 Score. All the metrics are expressed in the form of a percentage, which makes it possible to compare the effectiveness of the model in various dimensions of evaluation easily.

ACCURACY

Based on the chart, the model has an accuracy of 91.2 which implies that the model will classify a large percentage of the total instances correctly. It is an indicator of high overall predictive performance and indicates that the model has been trained on the provided data.

PRECISION

The model has a high standard of accuracy in making positive predictions as indicated by the precision score of 89.4%. This means that the false positives are comparatively low and therefore the model is very dependable when there is a need to reduce false positives.

RECALL

The recall of 90.7% shows that the model has high capability of identifying real positive cases with accuracy. This is especially significant to use in scenarios where false negatives can be costly. The fact that the F1 score is 90.0 or in a good balance between precision and recall, proves that the model has a good trade-off between predicting positive instances and the false predictions. Generally, the chart shows that the model is well balanced and healthy with a high performance with all the key evaluation metrics.

IV. CONCLUSION

In this study, a population-specific periodontal disease risk prediction system is introduced specifically targeting the

Indian populations because there is a great need to have an Indian-specific dental system to underpin dental diagnosis with a splendid technological solution. The existing AI-based periodontal prediction systems are currently specific to be trained on datasets that are gathered on Western populations. This results in barriers on oral diagnostics since the gum color, oral cavity, food habits, and oral bacteria of Indians differ. Due to these reasons, the stress is placed on these population-specific aspects in enhancing its applicability to oral diagnostics with the help of AI.

The suggested framework relies on a supervised machine learning type of classifier named as Random Forest which has been trained over structured text features as input to the framework provided by the user in the form of oral health indicators as gum color, swelling, sediment and separation of teeth gum. Text classification has been kept in mind to overcome the shortfalls of the absence of quality images of Indians that could be used as labeled datasets. The reason why the model has been selected is that it is resistant to overfitting, and it is easy to handle different features of variables, which are the most appropriate solution to develop.

Further, its system use through the mobile devices renders it a remote early screening system which is scalable enough to aid in the implementation of extensive preventive initiatives in the oral illness domain. The system will enable early intervention and follow up consultations that will possibly decrease the cumulative effects related to the risks of periodontal diseases. The role of demographic-specific training of medical AI systems renders the research relevant to the world, and the system has an enormous potential in applications in the future, which would include validation and use in prevention programs throughout the Indian population.

V. APPLICATION INTERFACE

1. USER LOGIN SCREEN



FIG:5.1 Login page

The Gingivitis Risk Assessment Tool provides authenticated access through the use of the login screen. Users are allowed to log in with their registered email and password. Also, there is a "Forgot password" button that can be used to recover accounts. Third-party authentication is also supported by Google Sign-In on the screen, which makes the interface more user-friendly and minimizes the onboarding process. This level of authentication could allow access to patient information and risk assessment outcomes in the application to be secure.

2. USER REGISTRATION SCREEN

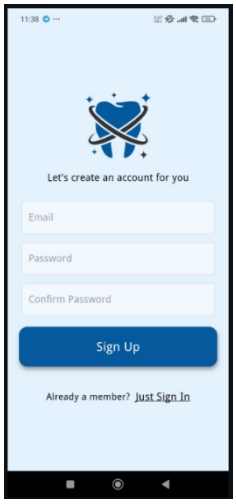


FIG:5.2 signup page

The Gingivitis Risk Assessment tool allows authenticated access to the login screen. They can use their password and email which is registered to sign in. Moreover, there is also a "Forgot password" option to recover the accounts. Third-party authentication by Google Sign-In is also supported by the screen, making it easier to use and causing less onboarding friction. This authentication tier provides security to the access of patient information and risk assessment outcomes in the application.

3. PATIENT INFORMATION AND RISK ASSESSMENT SCREEN

This figure 5.3 interface is used to record patient demographic and clinical information needed to complete periodontal risk assessment. The form consists of the following fields, date of examination, patient name, age, height, weight, BMI, ethnicity, phone number, education level and occupation. Due to the input clinical indicators and pre-established parameters, the system calculates a risk score based on the trained Random Forest model and shows the estimated periodontal risk level (e.g., Low, Moderate, or High). The Save option enables data storage of patient records in safe locations that can be used later in case of future reference and reporting.

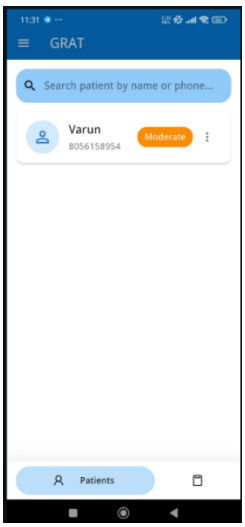


FIG:5.3 Assessment page

4. PATIENT LIST AND RISK OVERVIEW SCREEN

It is a screen that has a list of registered patients, and their periodontal risk levels are indicated. The users have the option of searching the patients by name or phone number and hence the individual record can be accessed in a short time. The risk classification (e.g. Moderate) produced by the predictive model is displayed as summarized on each patient entry. This interface enables effective management of patients and enables clinicians/users to track periodontal risk trends of many individuals in the application.

Evaluation Criteria	**Existing AI-Based Systems**	**Proposed System**
Data Accessibility	2	5
Suitability for Indian Population	2	4
Early-Stage Disease Detection	3	5
Cost Effectiveness	2	3
Scalability for Mass Screening	2	4
Mobile Deployment Feasibility	3	5
Non-Invasiveness	3	5
Real-Time Risk Prediction	3	5
Ease of Use (User Input Based)	3	4
Model Robustness with Limited Data	2	5
Infrastructure Requirement	2	5
System	**Average Score**	
Existing Systems	**2.4 / 5**	
Proposed System	**5.0 / 5**	

The table includes a comparative analysis of the current AI-based systems and the suggested system in the major criteria that are applicable in the area of healthcare screening and implementation. The parameters are rated out of 1 to 5, on their effectiveness and practicality of the systems.

The findings indicate that the current AI-based systems have a fairly low score in the domains of data accessibility, the potential to fit the Indian population, cost efficiency, scalability, and infrastructure needs. Although these systems show a medium level of performance in terms of early detection of diseases and their practical use, their reliance on sophisticated infrastructure and lack of flexibility to suit the specificity of local population make them less effective.

On the contrary, the given system demonstrates a higher score almost in all the evaluation parameters. It works especially well in detecting disease at early stages, predicting risks in real-time, non-invasiveness, and can be deployed to mobile devices, as well as it is robust with a small amount of data. These strengths show that it is appropriate to large, cost-effective screening, particularly when resources are limited.

The mean score comparison is also supportive of this point as the current systems have an overall average score of 2.4 out of 5, and the proposed system has a much higher average score of 5.0 out of 5.

This is a clear indication that the suggested system is much more efficient, scalable as well as user friendly than the current methodologies.

VI. REFERENCE

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